

eral-purpose PCB by using two connectors—one for the input and the other for the output. You can also use a DC-DC converter circuit in place of the linear regulator to avoid the use of transformer and also to reduce dis-

sipation. Finally, short-circuit protection is provided for all of the auxiliary outputs by clips, internal current limiter and thermal protection circuit. The specification of inductor L1 is shown in Fig. 2. It is a ferrite toroid core T-

18 with 22 turns of 27SWG enameled copper wire. ●

EFY note. During testing the output current was found to be about 220 mA only but still the circuit worked.

SPY CAMERA SOLAR POWER BOX

■ **TK. HAREENDRAN**

Battery life has always been a critical consideration for most electronic gadgets and equipment. When we use spy cameras, which normally function round-the-clock, they often run out of power within a few days.

disposable or rechargeable batteries, their frequent replacement or recharging is a cumbersome job. The unique solar power box described here serves an alternative solution to the problem.

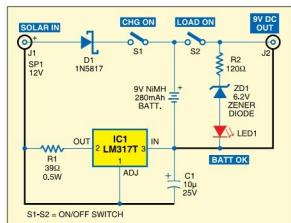
The circuit of the solar power box is simple. It contains a battery charger and a battery health indicator, besides a few other components. As shown in the circuit, DC supply available from the solar panel (SP1) is directly applied to the input of the circuit through a protection diode (D1). This diode is used to prevent the reverse current flow from the battery to the solar panel during night. Thus, D1 allows the current to flow from the solar panel to the battery only. Low-voltage

(pin 2) and ADJ (pin 1) terminals. Here a 9V, 280mAh rechargeable PP3 type Ni-MH battery (BATT) is used as reservoir. Normally, a charging current of about 10 per cent of ampere-hour rating is safe for the battery.

Resistor R1 (39-ohm, 0.5W), connected between pin 1 and 3 of IC1, limits the charging current to about 30 mA. DC output from the battery is available at output jack J2. Red LED (LED1) is used as a battery 'health' indicator. Switch S1 is used to start the charging while S2 is used for connecting the load. Note that suitable heat-sink should be used for the IC1.

The proper selection of solar panel is important but not critical. A miniature 12V type solar panel with a current output of about 100 mA can be used. Even if you have a solar panel with higher voltage rating, it will not create a problem as the circuit ensures that the charging current cannot exceed the predetermined value.

The circuit can be easily assembled on a general-purpose PCB and housed in a small plastic cabinet. ●



Many spy cameras (CCTV cameras) are powered by 9V PP3 type batteries that offer more energy than the regular 9V alkaline battery. Mini CCTV cameras also accept 6-12V DC supply from AC mains adaptor through the DC IN jack. AC mains adaptor for the camera increases the capacity of the 9V PP3 battery but is bulky and noisy. Whether using

age-drop type 1N5817 Schottky diode is perfect for the job.

At the heart of the circuit is an integrated current source, realised using a popular 3-pin adjustable voltage regulator LM317T (IC1). This IC is designed to adjust its internal resistance between the IN (pin 3) and OUT (pin 2) terminals to maintain a constant voltage of 1.25V between the OUT



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